

Preface

Why lagrange formalism works?

As we all know, a fast way to get a theory of field is to guess and write down a lagrangian density of the field. For example, a quite simple theory of (real) scalar is

$$\mathcal{L} = -\frac{1}{2}\partial_\mu\varphi\partial^\mu\varphi - \frac{1}{2}m^2\varphi^2 + V(\varphi)$$

where $V(\varphi)$ is the interaction potential. The direct advantage of this method is that the lagrangian formalism is manifestly Lorentz invariant. However, we can't get the obvious physical picture of *field* φ . And, the (canonical) quantization procedure may be not quite natural.

In the first part of this note, we will state some basic conditions that satisfy special relativity and quantum mechanics at the same time. Then we will find the way to describe quantum fields using lagrange formalism is effective and natural.

To begin with, let's review basic property of relativity and quantum mechanics. The frontier tells us that the structure of spacetime is determined by $4-d$ Minkowski space $(\mathbb{R}^4, g_{\mu\nu})$. And the latter one tells us that the state of a physical system is represented by a class of vectors that differ by a phase factor in a Hilbert space $\mathcal{H}$. Then the symmetry of spacetime should be represented by unitary (or anti-unitary) operators acting on $\mathcal{H}$ by the condition of *probability conservation*:

$$\langle\Psi|\Phi\rangle = \langle U(\Lambda)\Psi|U(\Lambda)\Phi\rangle$$

where $U(\Lambda)$ is the unitary representation of Lorentz transformation Λ . In the chapter 1, we will discuss the structure of these unitary representations and their relation with particles.

The other two basic conditions are *locality* and *causality*. Locality means that the interaction of fields happens at the same spacetime point. Causality means that the effect can't propagate faster than light. In chapter 2, we will introduce sactter matrix S which is unitary, then *the principle of cluster decomposition* determined by locality disscussed in chapter 3 with S will reveal the constraint on interaction of fields, i.e. the formalism of $V(\varphi)$.

Finally, we can observe the physical meaning of quantum field φ , and the lagrange quantity of field theory is self-consistent to describe quantum fields.

Ways to beautiful interactions and rigorous calculations

Ugly formalisms but essential results